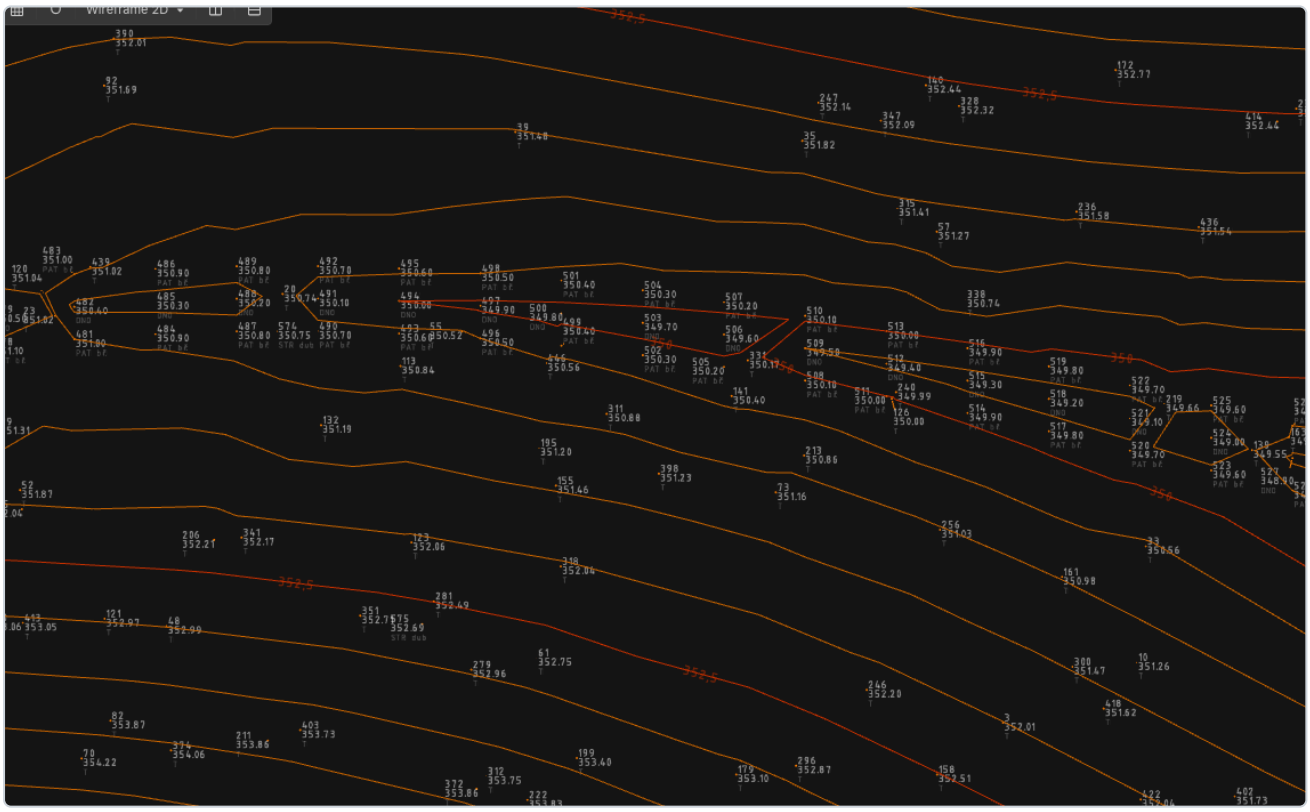
Terén

Open CAD Studio plugin: survey points, TIN surfaces, breaklines, contours

Short manual · version __VERZE__



1. What Terén does

Terén builds a digital terrain model from survey points directly in an Open CAD Studio drawing:

- **Points** – import from text files (PNEZD, PENZD, PYXZD, NEZ, ENZ…, the format is detected automatically), markers and labels by styles, point groups, manual points, survey check.
- **Surfaces** – triangulated network (TIN) from point groups, breaklines, boundaries and holes, points from drawing objects, manual triangle edits, automatic recalculation.
- **Display** – contours with labels, triangles, outline, slope arrows; 3D viewer.
- **Export** – surface to LandXML or DXF (3D faces), group points to a text file.

Everything is stored in the drawing (DWG/DXF) – the drawing can be passed on and edited without the plugin; the plugin then reads the surface again.

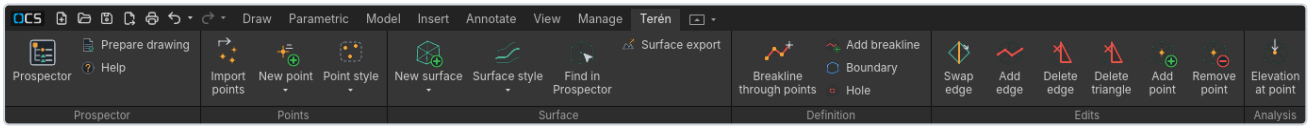
2. Installation

Plugin Manager: in Open CAD Studio open the Plugin Manager, find *Terén* and click Install. After a restart the **Terén** tab appears.

Manually (Windows): download `Teren-windows.zip` from the GitHub release, extract it and run `nainstalovat.bat`. The plugin is copied to `%APPDATA%\OpenCADStudio\plugins\cz.fluvio.teren`.

Each Terén release is built for one Open CAD Studio release – the Plugin Manager offers the right one; for a manual install pick the release from the table in the README (Terén 1.0.2 – 1.0.4 = Open CAD Studio 2026.45). The interface language (Czech / English) is set in Prospector › Settings.

3. Interface

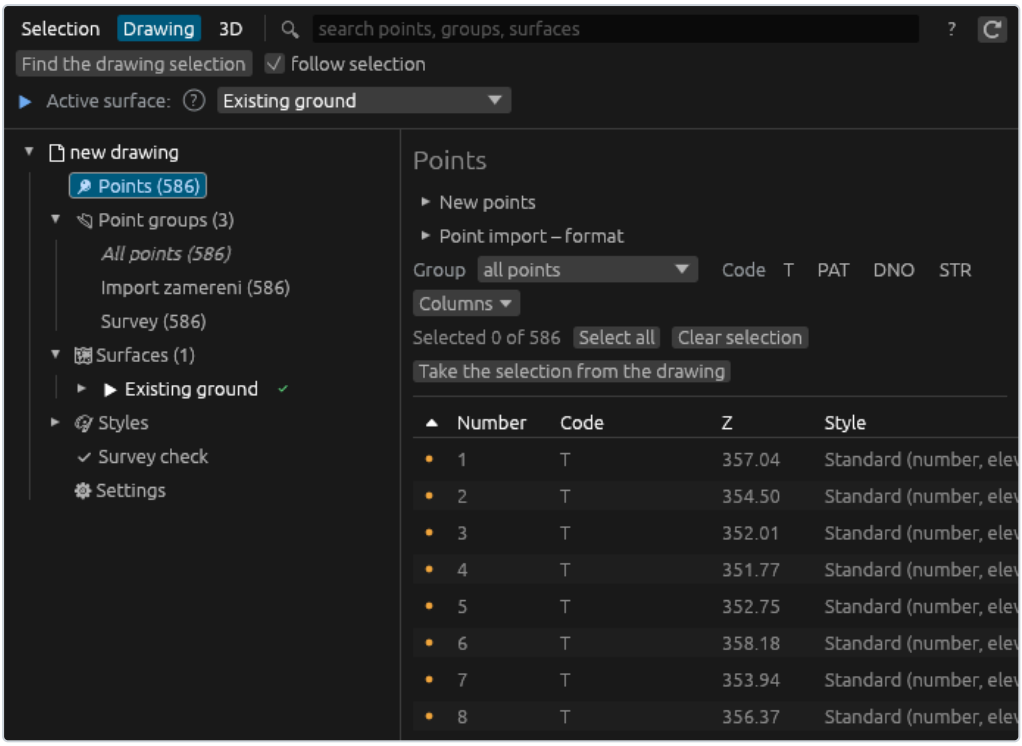


The Terén ribbon tab

Group	Contents
Prospector	Prospector window, Prepare drawing (layers, markers, template), Help
Points	Import points, New point (manual, on surface), Point style for selected points
Surface	New surface (existing ground, design from 3D lines, from clicked points, wizard), Surface style, Find in Prospector, Surface export
Definition	Breakline through points, Add breakline, Boundary, Hole
Edits	Swap / add / delete edge, delete triangle, add / remove point
Analysis	Elevation at point

The **Prospector** is a separate window with three modes:

- **Selection** – what is selected in the drawing (point, surface) and what you can do with it: edit elevation and number, style, groups, recalculation.
- **Drawing** – a tree of everything the plugin has in the drawing: points (table), point groups, surfaces, styles, survey check, settings. Right-click opens a menu of actions.
- **3D** – surface viewer.

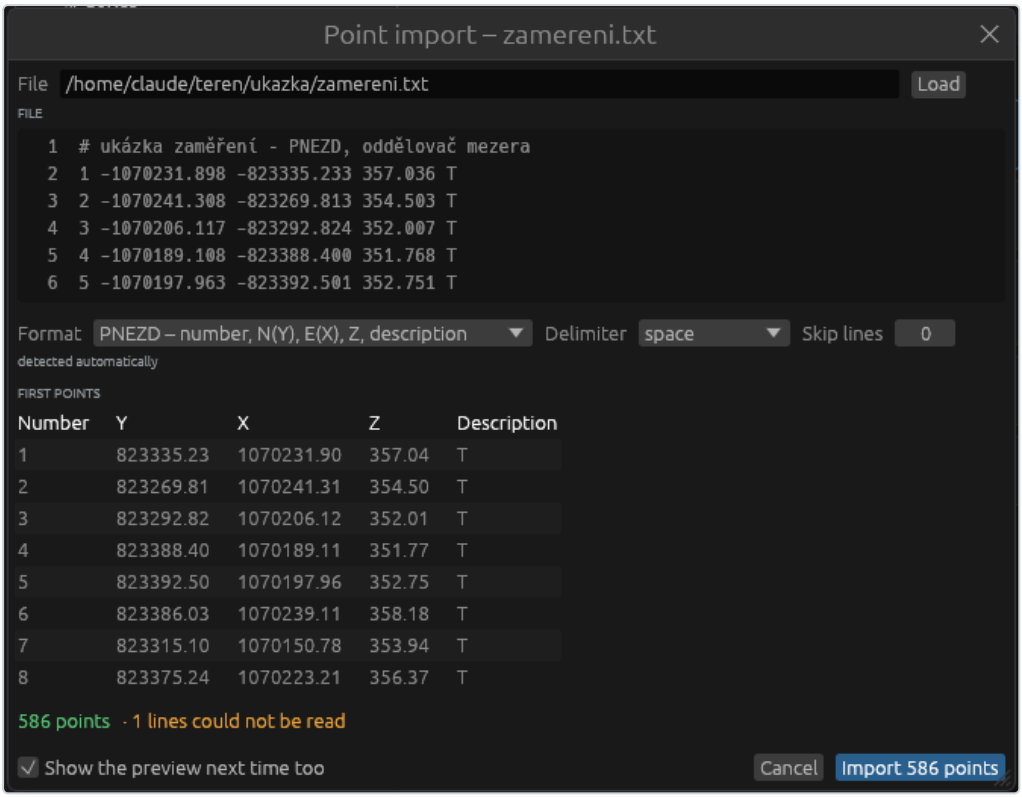


Prospector – Drawing mode, point table

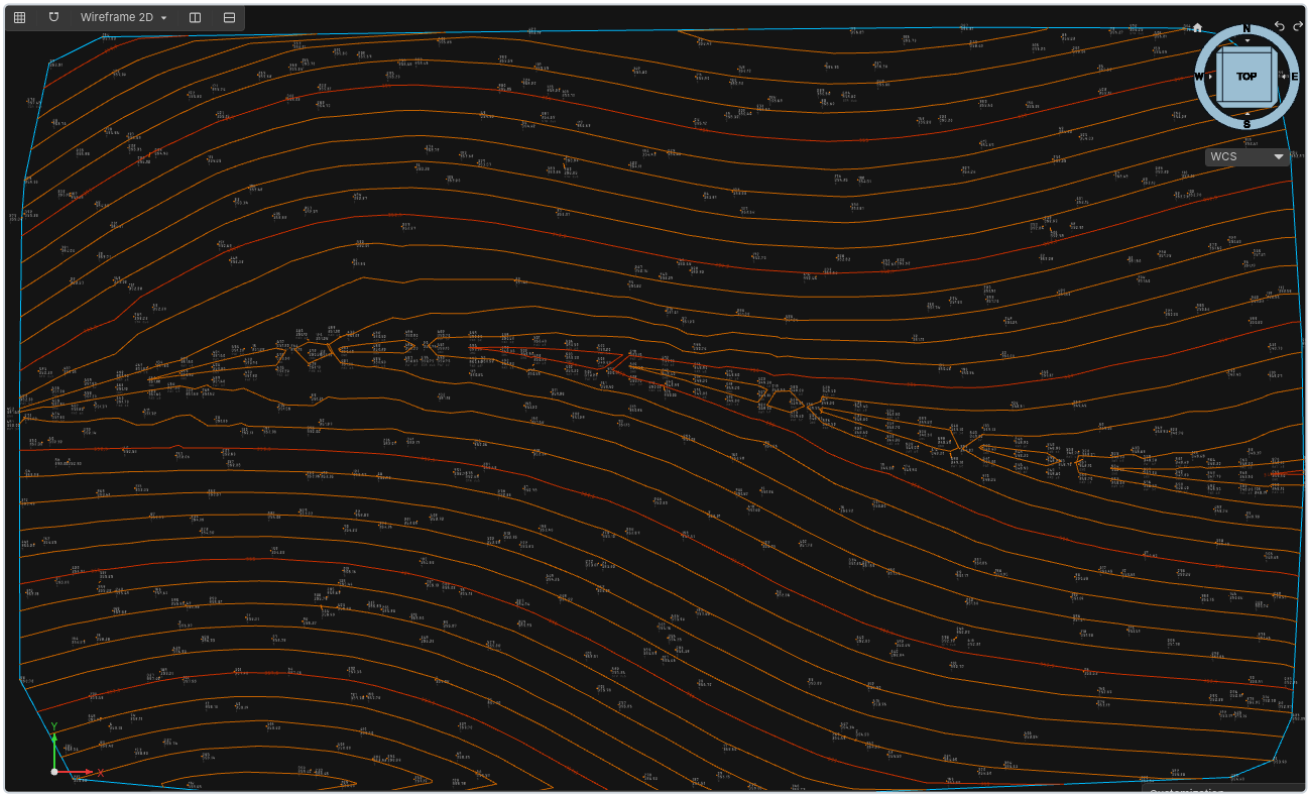
Help with workflows and a glossary is in the Prospector (**F1**); the ⓘ next to a term shows an explanation.

4. Ground from a survey – step by step

1. **Import points.** Terén › Import points, choose the file. The preview shows the first lines, the detected format and the first points; check the columns and click *Import*. S-JTSK points are drawn in negative drawing coordinates (positive Y, X in labels).
2. **New surface.** New surface › *Existing ground (from points)*. It takes the *Survey* group without STR codes (trees). The wizard (New surface › Wizard) lets you choose groups, max. edge length, style and contour interval.
3. **Breaklines.** *Breakline through points* – click survey points (slope edge, toe, top, ditch centreline); Enter = finish. Existing 3D polylines are added with *Add breakline*.
4. **Boundary.** Draw a closed polyline around the area, select it and click *Boundary*. *Hole* leaves out the inside of a polyline.
5. **Check and edit.** The *Triangles* style or the 3D tab shows the network. Swap or delete unsuitable edges with the Edits tools; each edit is stored with the surface and reapplied on recalculation.
6. **Contours.** Right-click the surface in the Prospector › Style (e.g. *Contours with labels*) and Contour interval (0.1 / 0.5 m …).



Import preview



Surface with contours

A surface with **automatic recalculation** recalculates itself when points, breaklines or boundaries change. Otherwise the Prospector marks it ⚠ out of date and you just click Recalculate. Each Terén action (import, recalculation, redraw) is undone in one **Undo** step (Ctrl+Z).

5. Points

Point styles

A style sets the marker (cross, circle, tree, manhole…) and the label (number, elevation, code). A manually set style applies, otherwise the style by code (the *for codes* field, e.g. STR → tree), otherwise the group's style, otherwise *Standard*. Sizes are in mm on paper and follow the drawing scale (annotation scale in the status bar, or Settings › Scale).

Custom marker: in the style editor choose *custom* – *drawing block* and select a block. The block's base point = point position, the block is scaled to the marker size; set the colour inside the block to ByBlock.

A point, its marker and labels stay together – clicking any part selects the whole point. **Moving a point:** grab the marker or a label and move it (with a grip or the Move command) – the whole point moves, its elevation stays and the surface is recalculated. With *moving a point label moves only the label* turned on in Settings, a moved label stays aside (with a leader line) and the point stays put; *Reset labels* in the point table brings it back. A copied marker or label is deleted. Overlapping labels are moved aside with a leader line (can be turned off in Settings).

Moving points by grip, Terén point panel, cursor menu

Move points (Terén tab › Points; the button shows *on / off*, switch with its arrow) turns on custom grips: a circle appears on every point. Grab it and drag – the marker and labels move with the point; on release the point moves (its elevation stays) in one undo step. Click *Move points* again to turn the grips off (while they are on, a click on a point's centre starts a move instead of selecting it).

Point properties opens the *Terén point* panel in the dock: after selecting a point it shows the number, elevation, code and style. Edit the fields and click *Save*. With several points selected only the changed fields are saved (the number only for a single point).

The **cursor menu** (quad; turned on in Open CAD Studio with F12 or the QUAD field in the status bar) has a *Terén* tab for points and surfaces: point properties, move points, add point to surface, recalculate surface, Prospector.

Manual points and editing elevation

New point › Manual point: click in the drawing or enter x,y; a window asks for the elevation (suggested from the surface or the previous point) and the code. *Point on surface* takes the elevation of the active surface. Change the elevation, number and code of an existing point directly in the point table: double-click a cell, type, Enter saves and moves to the next point (Esc cancels). For several selected points at once, set or add an elevation and set a code in the row above the table.

Point groups

A group selects points by rules (code, numbers, import file), with exclusions and manually added / removed points. The preview shows in colour which points belong. A surface is built from groups – a change of a group recalculates the surface or marks it out of date.

Survey check

Prospector › Survey check finds points whose elevation differs markedly from their surroundings (typo, point on a car). Suspicious points are circled on the TER_KONTROLA layer; one click leaves them out of the surface.

6. Surfaces

A surface is built from point groups, drawing objects (points, blocks, lines, 3D faces – e.g. a surveyor's drawing), breaklines, boundaries and holes, and manual edits. All are listed in the

Prospector at the surface › *What the surface is built from*; each item can be disabled, shown or removed.

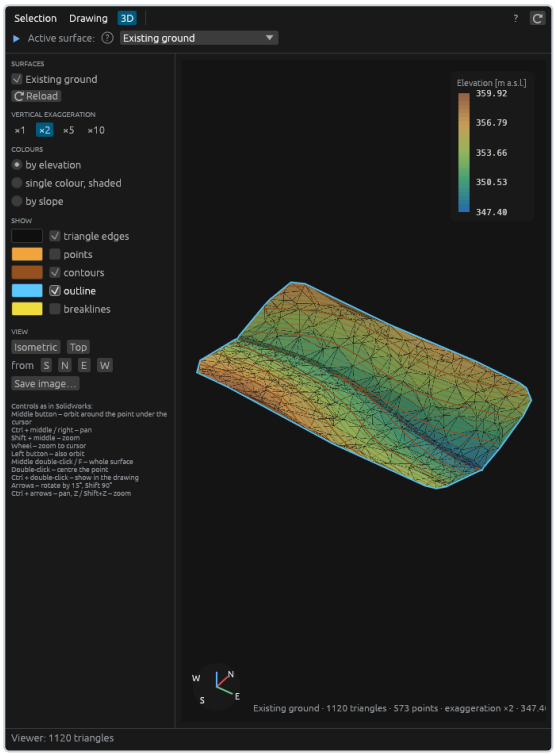
The **active surface** (top row of the Prospector, double-click in the tree) is used by the Edits tools, Elevation at point and Point on surface.

The **surface style** sets what is drawn: triangles, contours (minor / major, colours, labels), outline, slope arrows. Edit styles in the Prospector under Styles; a built-in style can always be restored to its original settings.

In the drawing a surface is a block on its own layers; Settings › *Surface cannot be selected by clicking* lets clicks pass through to the points below.

7. 3D viewer

Colours by elevation, slope or a single colour, vertical exaggeration ×1 to ×10, contours, outline, breaklines, points and triangle edges – pick the colour of each element with the swatch next to it. Controls as in SolidWorks: the middle button orbits around the point under the cursor, Ctrl + middle pans, the wheel zooms, F shows the whole surface. Save image writes a PNG to the drawing folder.



8. Export

- **Surface export** (ribbon or right-click a surface): **LandXML 1.2** – exactly the same triangles, survey points and outline, readable by most design programs; **DXF** – each triangle as a 3D face.
- **Point export** of a group to a text file (PNEZD, PYXZD…, delimiter, UTF-8 or Windows-1250 encoding).

9. Settings and drawing template

Prospector › Settings: language (Czech / English, takes effect after Open CAD Studio is restarted), scale, label offset, import preview, excluded codes for a new surface, breakline snap, survey check threshold.

Drawing template: *Save this drawing as the template* stores the styles and scale in the user profile; *Prepare drawing* (ribbon) then sets up layers, point markers, scale and styles in a new drawing with one click.

10. Tips and troubleshooting

Problem	Solution
No points appeared after import	Check the format and column order in the preview (S-JTSK: positive Y and X in the PYXZD format).
Long triangles across the edge of the surface	Set the Max. edge length or add a boundary.
Triangles cross a slope edge	Add a breakline (Breakline through points).
Trees are in the ground	Code STR is in the surface's *Exclude codes*; add other codes that do not belong to the ground.
The surface shows ⚠ out of date	Click Recalculate, or turn on automatic recalculation.
Labels are too large / small	Change the scale (status bar or Settings) – labels are redrawn.

Terén · Ing. Pavel Janouš · GPL-3.0 licence · source code and issue reports: GitHub